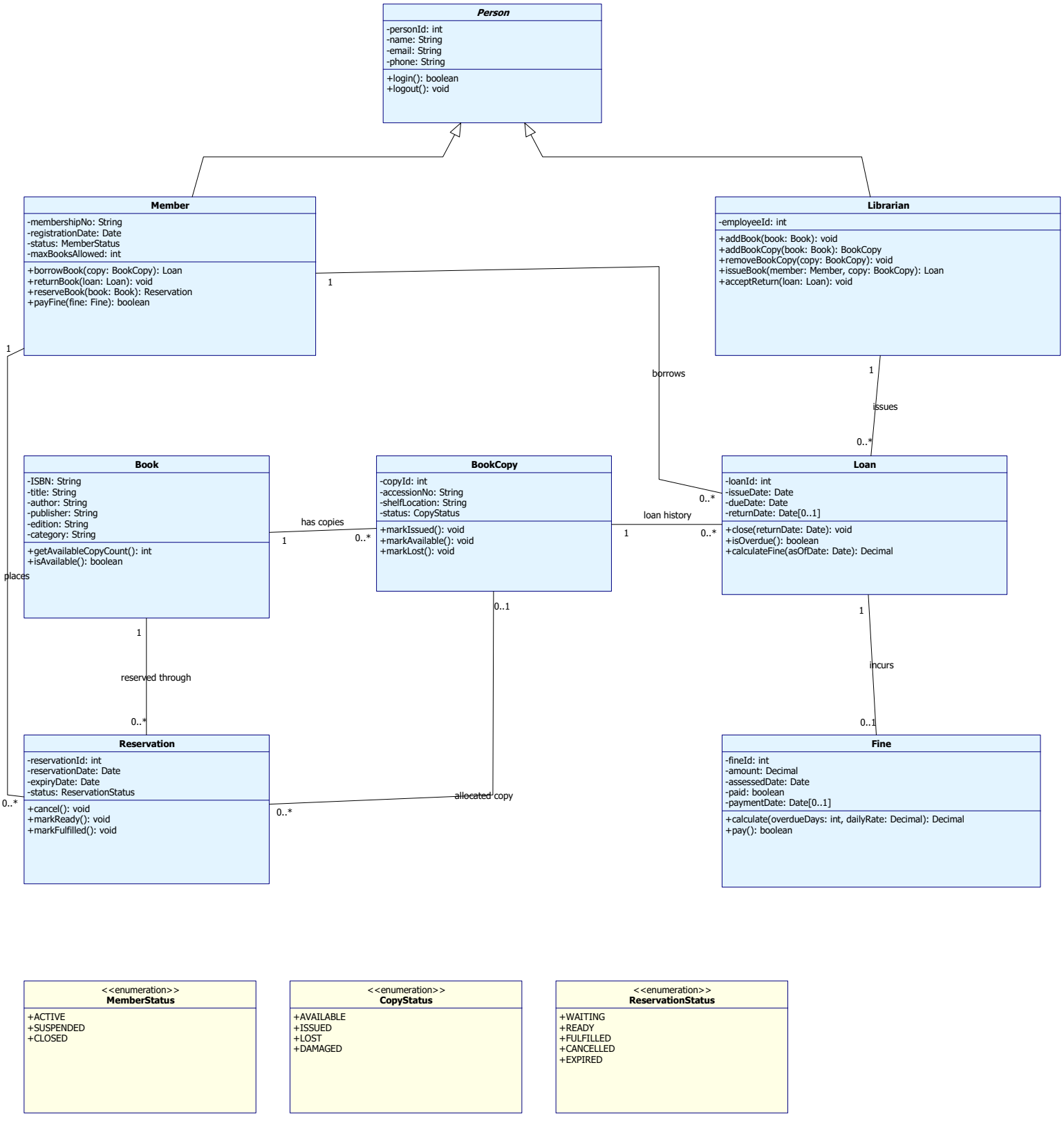

Person
-personId: int
-name: String
-email: String
-phone: String
+login(): boolean
+logout(): void
Member
-membershipNo: String
-registrationDate: Date
-status: MemberStatus
-maxBooksAllowed: int
+borrowBook(copy: BookCopy): Loan
+returnBook(loan: Loan): void
+reserveBook(book: Book): Reservation
+payFine(fine: Fine): boolean
Librarian
-employeeId: int
+addBook(book: Book): void
+addBookCopy(book: Book): BookCopy
+removeBookCopy(copy: BookCopy): void
+issueBook(member: Member, copy: BookCopy): Loan
+acceptReturn(loan: Loan): void
borrows
issues
Book
-ISBN: String
-title: String
-author: String
-publisher: String
-edition: String
-category: String
+getAvailableCopyCount(): int
+isAvailable(): boolean
has copies
BookCopy
-copyId: int
-accessionNo: String
-shelfLocation: String
-status: CopyStatus
+markIssued(): void
+markAvailable(): void
+markLost(): void
loan history
Loan
-loanId: int
-issueDate: Date
-dueDate: Date
-returnDate: Date[0..1]
+close(returnDate: Date): void
+isOverdue(): boolean
+calculateFine(asOfDate: Date): Decimal
places
reserved through
incurs
allocated copy
Reservation
-reservationId: int
-reservationDate: Date
-expiryDate: Date
-status: ReservationStatus
+cancel(): void
+markReady(): void
+markFulfilled(): void
Fine
-fineId: int
-amount: Decimal
-assessedDate: Date
-paid: boolean
-paymentDate: Date[0..1]
+calculate(overdueDays: int, dailyRate: Decimal): Decimal
+pay(): boolean
<<enumeration>>
MemberStatus
+ACTIVE
+SUSPENDED
+CLOSED
<<enumeration>>
CopyStatus
+AVAILABLE
+ISSUED
+LOST
+DAMAGED
<<enumeration>>
ReservationStatus
+WAITING
+READY
+FULFILLED
+CANCELLED
+EXPIRED